

Lab report, part B, number 3 - 2025.

This lab will require Google Collab

The objective of this lab is to write a function to calculate the required weights for a Hopfield network in order to store a given pattern. The input to the function will be a vector of Neuron values as the pattern to remember. The return should be a matrix of weights.

Hint. Remember that the weights can be calculated as the outer product of the pattern with itself.

Test your function on the following 3x3 pattern



submission for your report:

-Show the results of your google collab including

1. The function code
2. The values of the input vector
3. The values of the resulting matrix
4. Answer the question: Why should we zero the diagonal of the weight matrix? (1 sentence max)